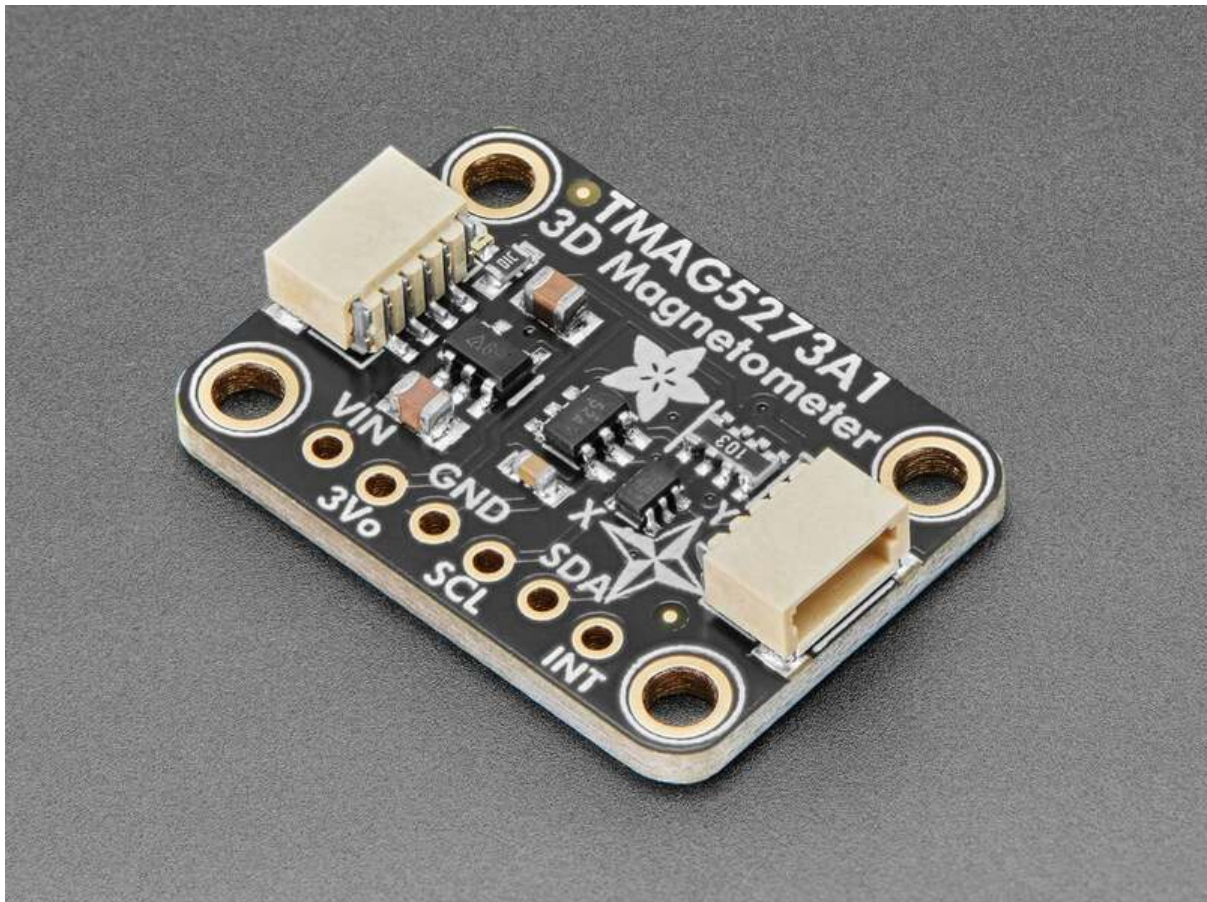




Adafruit TMAG5273 3D Hall Effect Magnetometer Breakout

Created by Liz Clark



<https://learn.adafruit.com/adafruit-tmag5273-3d-hall-effect-magnetometer-breakout>

Last updated on 2026-03-25 09:07:14 PM UTC

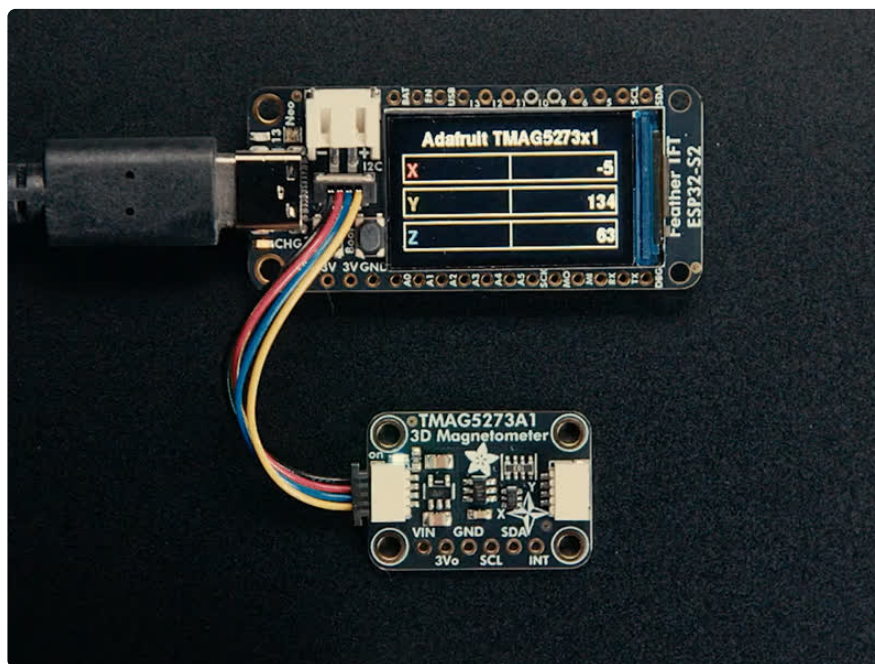
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Overview

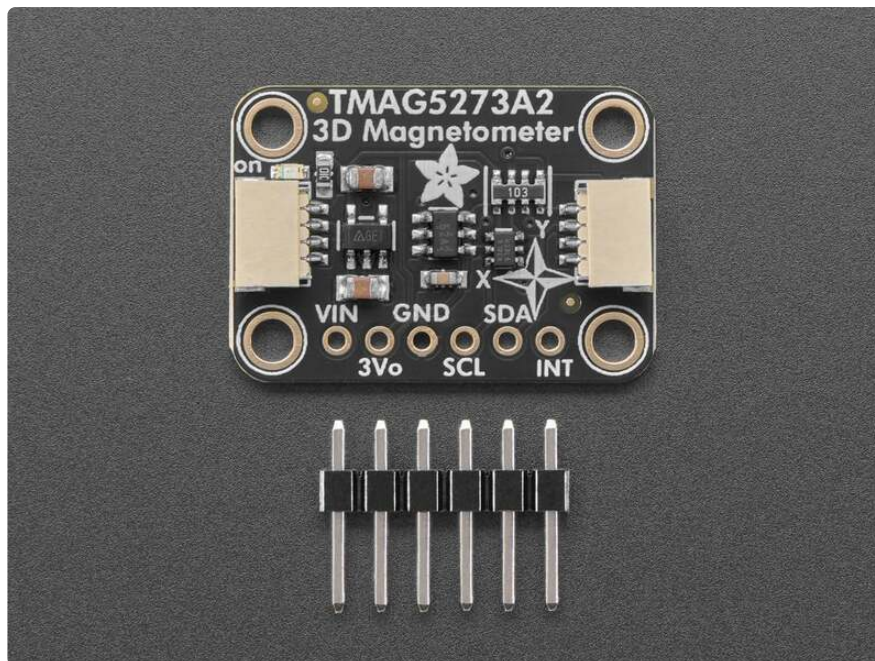


guide is for both the A1 and A2 variants. Both use the exact same
ware and pinout. The A1 type has a $\pm 40\text{mT}$ / $\pm 80\text{mT}$ range and the A2 has
der $\pm 133\text{mT}$ / $\pm 266\text{mT}$ range.



sensor is meant for sensing rotation or motion of an object nearby with a
net embedded inside. It is not meant for measuring the magnetic field of
earth.

The **Adafruit TMAG5273 3D Hall Effect Magnetometer Breakout** is a low-power linear 3D Hall-effect sensor designed for a wide range of industrial and personal electronics applications. It integrates three independent Hall-effect magnetometer sensors in the X, Y, and Z axes for full 3D orientation detection. The built-in 12-bit ADC can read the signal from each axis for read-out over I2C, it even has a thermometer built in for basic temperature reading!



An integrated angle calculation engine (CORDIC) provides full 360° angular position information for both on-axis and off-axis angle measurements. The angle calculation is performed using two user-selected magnetic axes over I2C. The TMAG also features magnetic gain and offset correction - note you'll have to manage the values and write them into the chip on boot-up.



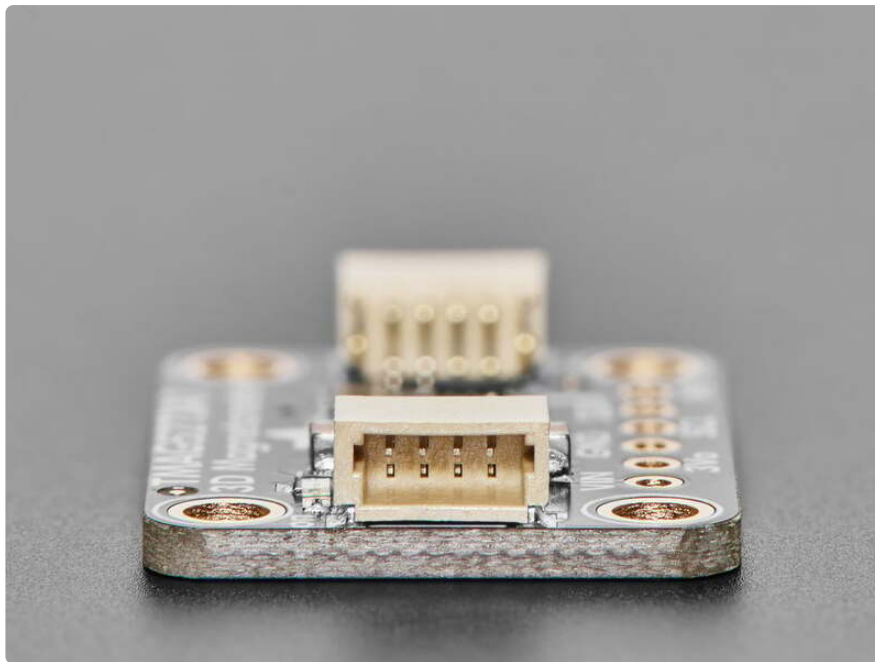
3D Hall effect sensors are great for sensing rotation or motion of an object nearby with a magnet embedded inside: often used for magnetic rotary encoders or joysticks

that don't have mechanical switches and thus are more reliable and don't fail from wear.

This sensor comes in two flavors, the **A1** type is $\pm 40\text{mT}$ / $\pm 80\text{mT}$ range (you can pick either range), and the **A2** has a wider $\pm 133\text{mT}$ / $\pm 266\text{mT}$ range. Both use the exact same firmware and pinout - the firmware we've written can detect which one you have plugged in and automatically scale the readings as necessary. This isn't a great sensor for measuring the magnetic field of the earth, much better for magnets of the ferric or rare-earth persuasion.



To make life easier so you can focus on your important work, we've taken the TMAG5273 and put it onto a breakout PCB along with support circuitry to let you use this little wonder with 3.3V (Feather/Raspberry Pi) or 5V (Arduino/ Metro328) logic levels. Additionally, since it speaks I2C, you can easily connect it up with two wires (plus power and ground!). We've even included [SparkFun qwiic \(https://adafru.it/Fpw\)](https://adafru.it/Fpw) compatible [STEMMA QT \(https://adafru.it/Ft4\)](https://adafru.it/Ft4) connectors for the I2C bus so **you don't even need to solder!** Just wire up to your favorite micro and you can use our CircuitPython/Python or [Arduino drivers to easily interface with the TMAG \(https://adafru.it/1aBZ\)](https://adafru.it/1aBZ) and get magnetic measurements ASAP.

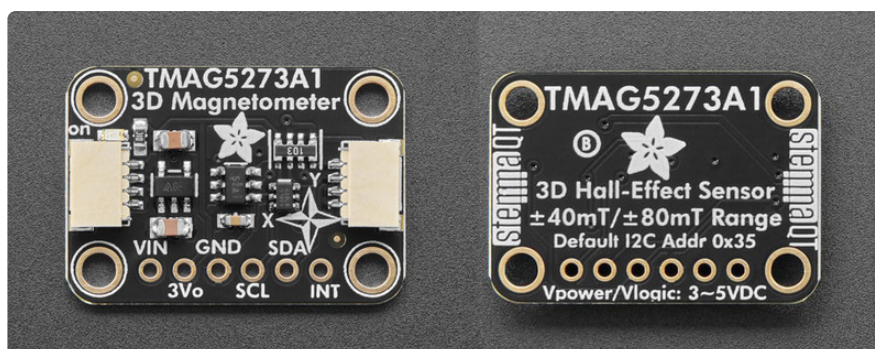


It's fully assembled and tested. Comes with a bit of 0.1" standard header in case you want to use it with a breadboard or perfboard. Four 2.5mm (0.1") mounting holes for easy attachment.

Pinouts



guide is for both the A1 and A2 variants. Both use the exact same ware and pinout. The A1 type has a $\pm 40\text{mT}$ / $\pm 80\text{mT}$ range and the A2 has der $\pm 133\text{mT}$ / $\pm 266\text{mT}$ range.



The default I2C address is **0x35**.

Power Pins

- **VIN** - this is the power pin. To power the board, give it the same power as the logic level of your microcontroller - e.g. for a 5V microcontroller like Arduino, use 5V. For a 3.3V microcontroller, use 3.3V.
- **3Vo** - this is the 3.3V output from the voltage regulator, you can grab up to 100mA from this if you like.
- **GND** - common ground for power and logic.

I2C Logic Pins

- **SCL** - I2C clock pin, connect to your microcontroller I2C clock line. This pin is level shifted so you can use 3-5V logic, and there's a **10K pullup** on this pin.
- **SDA** - I2C data pin, connect to your microcontroller I2C data line. This pin is level shifted so you can use 3-5V logic, and there's a **10K pullup** on this pin.
- **STEMMA QT** (<https://adafru.it/Ft4>) - These connectors allow you to connect to dev boards with **STEMMA QT** connectors or to other things with [various associated accessories](https://adafru.it/Ft6) (<https://adafru.it/Ft6>).

Interrupt Pin

- **INT** - This is the interrupt pin. It can be configured as an input or an output.

Power LED

- **Power LED** - In the upper left corner, above the STEMMA QT connector, on the front of the board, is the power LED, labeled **on**. It is a green LED.

CircuitPython and Python

It's easy to use the **TMAG5273** with Python or CircuitPython, and the [Adafruit_CircuitPython_TMAG5273](https://adafru.it/1aC0) (<https://adafru.it/1aC0>) module. This module allows you to easily write Python code that allows you to read the **TMAG5273** magnetometer. You can use this magnetometer with any CircuitPython microcontroller board or with a computer that has GPIO and Python [thanks to Adafruit_Blinka, our CircuitPython-for-Python compatibility library](https://adafru.it/BSN) (<https://adafru.it/BSN>).



sensor is meant for sensing rotation or motion of an object nearby with a net embedded inside. It is not meant for measuring the magnetic field of earth.



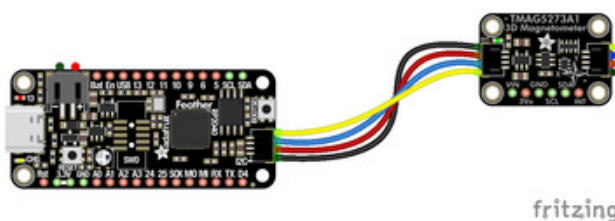
High-strength 'rare earth' magnet

Yow! These things are super powerful. .47" diameter and .18" thick discs. Great for use with your SpokePOV Kit. If you have an aluminium frame, use tape/foam sticky tape to...

<https://www.adafruit.com/product/9>

CircuitPython Microcontroller Wiring

First, wire up an TMAG5273 to your board exactly as shown below. Here's an example of wiring a Feather RP2040 to the TMAG5273 with I2C using one of the handy [STEMMA QT](https://adafru.it/Ft4) (<https://adafru.it/Ft4>) connectors:



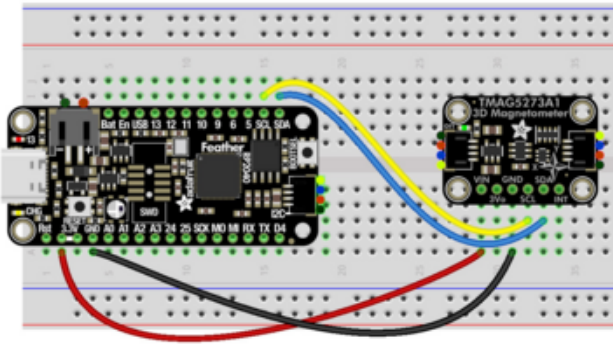
Board STEMMA 3V to breakout
STEMMA VIN (red wire)

Board STEMMA GND to breakout
STEMMA GND (black wire)

Board STEMMA SCL to breakout STEMMA
SCL (yellow wire)

Board STEMMA SDA to breakout
STEMMA SDA (blue wire)

You can also use standard **0.100"** pitch headers to wire it up on a breadboard:

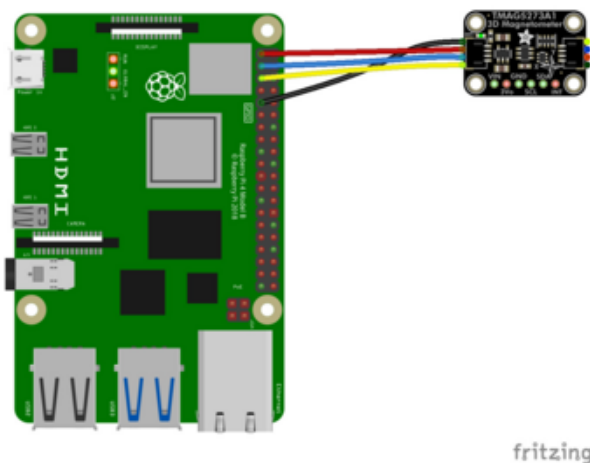


Board 3V to breakout VIN (red wire)
 Board GND to breakout GND (black wire)
 Board SCL to breakout SCL (yellow wire)
 Board SDA to breakout SDA (blue wire)

Python Computer Wiring

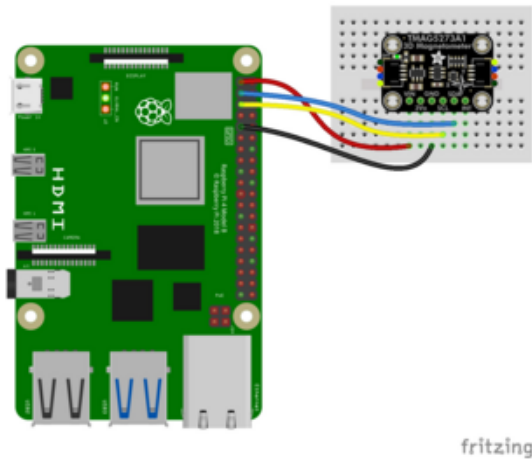
Since there's dozens of Linux computers/boards you can use, below shows wiring for Raspberry Pi. For other platforms, [please visit the guide for CircuitPython on Linux to see whether your platform is supported](https://adafruit.it/BSN) (<https://adafruit.it/BSN>).

Here's the Raspberry Pi wired to the magnetometer using I2C and a [STEMMA QT](https://adafruit.it/Ft4) (<https://adafruit.it/Ft4>) connector:



Pi 3V to breakout STEMMA VIN (red wire)
 Pi GND to breakout STEMMA GND (black wire)
 Pi SCL to breakout STEMMA SCL (yellow wire)
 Pi SDA to breakout STEMMA SDA (blue wire)

Finally, here is an example of how to wire up a Raspberry Pi to the magnetometer using a solderless breadboard:



Pi 3V to breakout VIN (red wire)
Pi GND to breakout GND (black wire)
Pi SCL to breakout SCL (yellow wire)
Pi SDA to breakout SDA (blue wire)

Python Installation of TMAG5273 Library

You'll need to install the **Adafruit_Blinka** library that provides the CircuitPython support in Python. This may also require enabling I2C on your platform and verifying you are running Python 3. [Since each platform is a little different, and Linux changes often, please visit the CircuitPython on Linux guide to get your computer ready \(https://adafru.it/BSN\)](https://adafru.it/BSN)!

Once that's done, from your command line run the following command:

- `pip3 install adafruit-circuitpython-tmag5273`

If your default Python is version 3, you may need to run `pip` instead. Make sure you aren't trying to use CircuitPython on Python 2.x, it isn't supported!

CircuitPython Usage

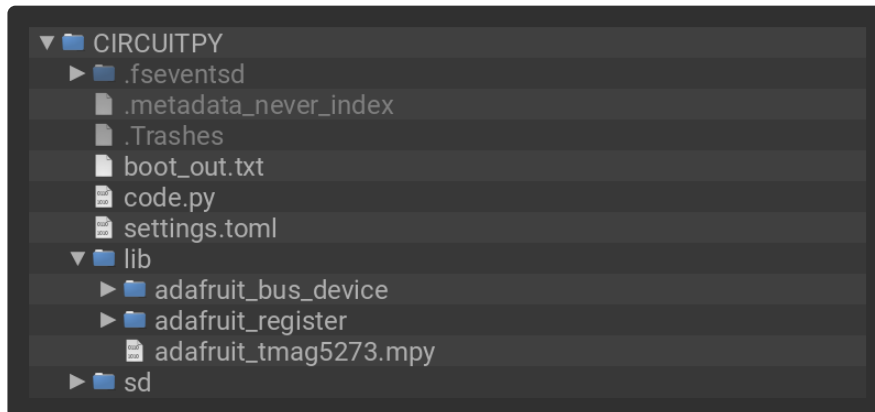
To use with CircuitPython, you need to first install the TMAG5273 library and its dependencies into the **lib** folder on your **CIRCUITPY** drive. Then you need to update **code.py** with the example script.

Thankfully, we can do this in one go. In the example below, click the **Download Project Bundle** button below to download the necessary libraries and the **code.py** file in a zip file. Extract the contents of the zip file, and copy the **entire lib folder** and the **code.py** file to your **CIRCUITPY** drive.

Your **CIRCUITPY/lib** folder should contain the following folders and file:

- **adafruit_bus_device/**
- **adafruit_register/**

- adafruit_tmag5273.mpy



Python Usage

Once you have the library `pip3` installed on your computer, copy or download the following example to your computer, and run the following, replacing `code.py` with whatever you named the file:

```
python3 code.py
```

Example Code

If running CircuitPython: Once everything is saved to the **CIRCUITPY** drive, [connect to the serial console \(https://adafru.it/Bec\)](https://adafru.it/Bec) to see the data printed out!

If running Python: The console output will appear wherever you are running Python.

```
# SPDX-FileCopyrightText: Copyright (c) 2026 Liz Clark for Adafruit Industries
#
# SPDX-License-Identifier: MIT

import time
import board
import adafruit_tmag5273

i2c = board.I2C()
sensor = adafruit_tmag5273.TMAG5273(i2c)

while True:
    mag_x, mag_y, mag_z = sensor.magnetic
    temp = sensor.temperature
    print(f"X: {mag_x:.2f} uT  Y: {mag_y:.2f} uT  Z: {mag_z:.2f} uT  Temp:
{temp:.1f} °C")
    time.sleep(1)
```

In the example, the temperature sensor is instantiated on I2C. Then, in the loop, the X, Y and Z readings from the magnetometer, along with temperature readings, are printed to the serial console every second. This sensor is meant for sensing rotation or motion of an object nearby with a magnet embedded inside. It is not meant for measuring the magnetic field of the earth.

```
X: 2240.48 uT Y: 1850.83 uT Z: -24.35 uT Temp: 29.9 °C
X: 3466.25 uT Y: 3466.25 uT Z: 170.47 uT Temp: 29.9 °C
X: 4140.01 uT Y: -811.77 uT Z: -4570.25 uT Temp: 29.9 °C
X: 3799.07 uT Y: 1777.77 uT Z: -4269.90 uT Temp: 29.8 °C
X: 2516.48 uT Y: -1315.06 uT Z: 1104.00 uT Temp: 29.9 °C
X: 2662.60 uT Y: -2824.95 uT Z: -795.53 uT Temp: 30.1 °C
X: 1664.12 uT Y: -1858.95 uT Z: 1923.89 uT Temp: 30.1 °C
X: 1030.94 uT Y: -495.18 uT Z: 3328.25 uT Temp: 30.0 °C
X: 3888.37 uT Y: -292.24 uT Z: -1168.95 uT Temp: 29.9 °C
X: 3685.42 uT Y: 925.42 uT Z: -1363.77 uT Temp: 29.9 °C
X: -97.41 uT Y: -308.47 uT Z: -235.41 uT Temp: 29.9 °C
```

Python Docs

[Python Docs \(https://adafru.it/1aBJ\)](https://adafru.it/1aBJ)

Arduino

Using the TMAG5273 magnetometer with Arduino involves wiring up the magnetometer to your Arduino-compatible microcontroller, installing the [Adafruit_TMAG5273 \(https://adafru.it/1aBZ\)](https://adafru.it/1aBZ) library and running the provided example code.



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High-strength 'rare earth' magnet

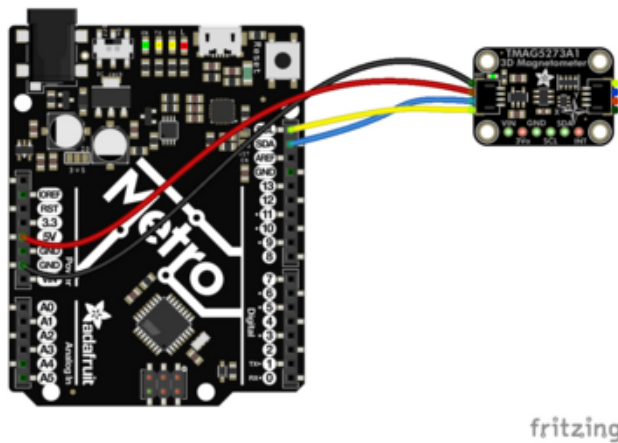
Yow! These things are super powerful. .47" diameter and .18" thick discs. Great for use with your SpokePOV Kit. If you have an aluminium frame, use tape/foam sticky tape to...

<https://www.adafruit.com/product/9>

Wiring

Wire as shown for a **5V** board like an Uno. If you are using a **3V** board, like an Adafruit Feather, wire the board's 3V pin to the TMAG5273 VIN.

Here is an Adafruit Metro wired up to the TMAG5273 using the STEMMA QT connector:



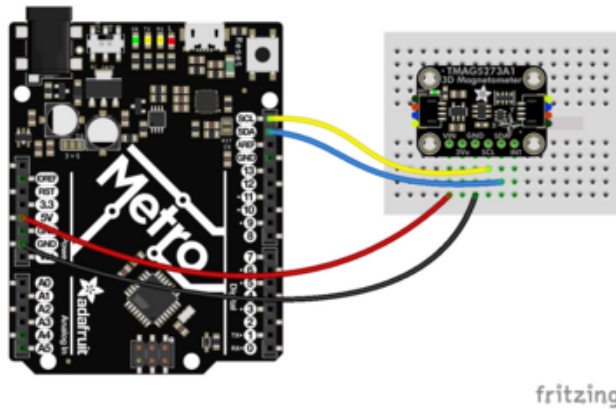
Board 5V to breakout STEMMA VIN (red wire)

Board GND to breakout STEMMA GND (black wire)

Board SCL to breakout STEMMA SCL (yellow wire)

Board SDA to breakout STEMMA SDA (blue wire)

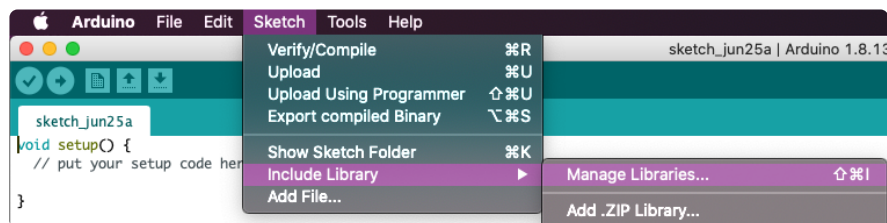
Here is an Adafruit Metro wired up using a solderless breadboard:



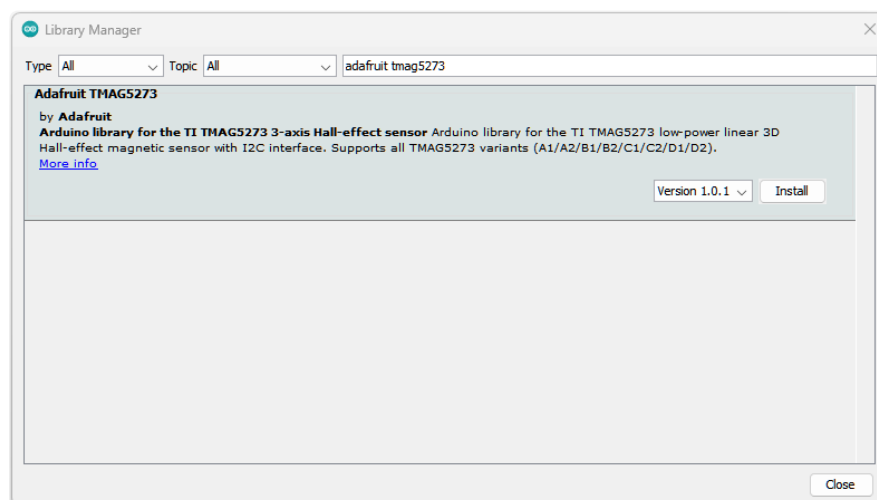
Board 5V to breakout VIN (red wire)
 Board GND to breakout GND (black wire)
 Board SCL to breakout SCL (yellow wire)
 Board SDA to breakout SDA (blue wire)

Library Installation

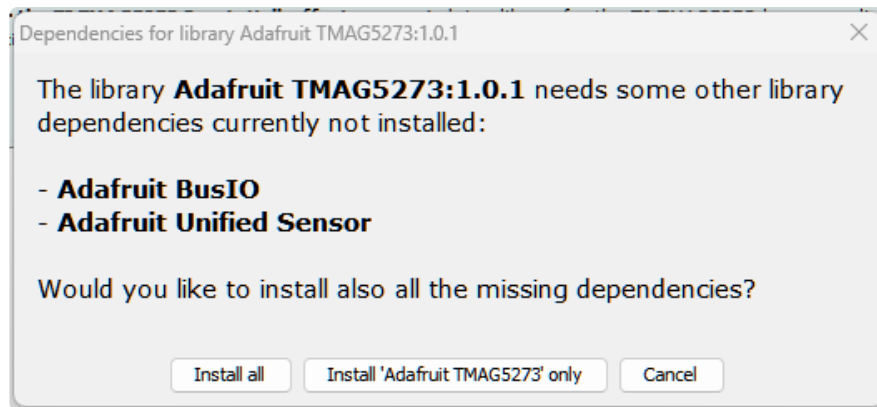
You can install the **Adafruit TMA5273** library for Arduino using the Library Manager in the Arduino IDE.



Click the **Manage Libraries ...** menu item, search for **Adafruit TMA5273**, and select the **Adafruit TMA5273** library:



If asked about dependencies, click **"Install all"**.



If the "Dependencies" window does not come up, then you already have the dependencies installed.



If the dependencies are already installed, you must make sure you update them through the Arduino Library Manager before loading the example!

Example Code

```
/*!
 * @file simpletest.ino
 * @brief Simple test for the Adafruit TMAG5273 3-axis Hall-effect sensor
 *
 * Reads magnetic field (X, Y, Z in uT) and temperature, prints to serial.
 *
 * Written by Limor 'ladyada' Fried with assistance from Claude Code
 * MIT license
 */

#include <Adafruit_TMAG5273.h>

Adafruit_TMAG5273 tmag;

void setup() {
  Serial.begin(115200);
  while (!Serial) {
    delay(10);
  }

  Serial.println(F("Adafruit TMAG5273 Simple Test"));

  if (!tmag.begin()) {
    Serial.println(F("Failed to find TMAG5273 sensor!"));
    while (1) {
      delay(10);
    }
  }
  Serial.println(F("TMAG5273 found!"));
```

```

Serial.print(F("Manufacturer ID: 0x"));
Serial.println(tmag.getManufacturerID(), HEX);
Serial.print(F("Device ID: 0x"));
Serial.println(tmag.getDeviceID(), HEX);
bool is_x2 = (tmag.getDeviceID() & 0x03) == 0x02;
Serial.print(F("Variant: "));
Serial.println(is_x2 ? F("x2 (+/-133/266 mT)") : F("x1 (+/-40/80 mT)"));
}

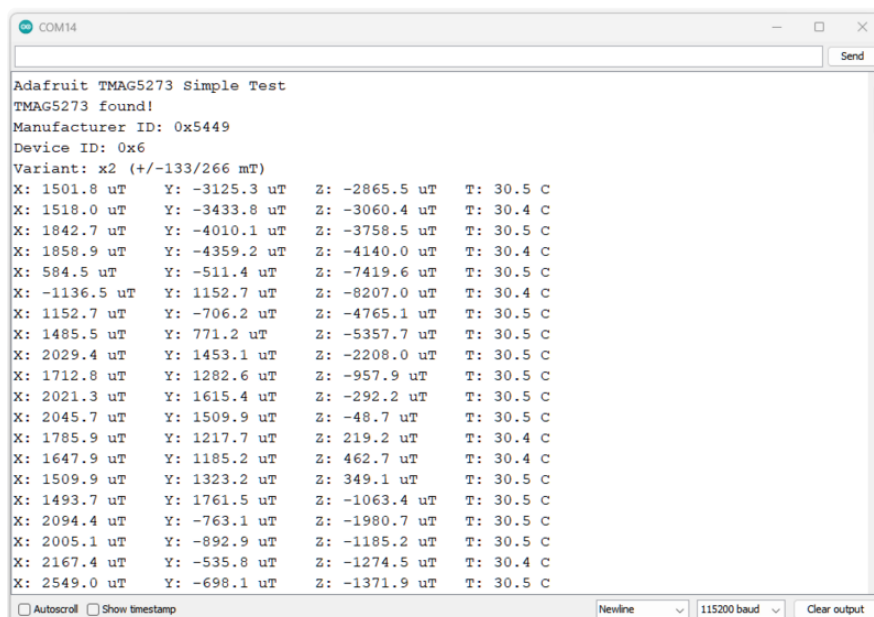
void loop() {
  float x = tmag.readMagneticX();
  float y = tmag.readMagneticY();
  float z = tmag.readMagneticZ();
  float temp = tmag.getTemperature();

  Serial.print(F("X: "));
  Serial.print(x, 1);
  Serial.print(F(" uT\tY: "));
  Serial.print(y, 1);
  Serial.print(F(" uT\tZ: "));
  Serial.print(z, 1);
  Serial.print(F(" uT\tT: "));
  Serial.print(temp, 1);
  Serial.println(F(" C"));

  delay(100);
}

```

Upload the sketch to your board and open up the Serial Monitor (**Tools -> Serial Monitor**) at 115200 baud. You should see that the sketch has found your connected TMAG5273 sensor. It will also print out the variant type (A1 or A2). Then, you'll see the X, Y and Z readings along with the temperature reading printed to the Serial Monitor every second. This sensor is meant for sensing rotation or motion of an object nearby with a magnet embedded inside. It is not meant for measuring the magnetic field of the earth.



Arduino Docs

[Arduino Docs \(https://adafru.it/1aBI\)](https://adafru.it/1aBI)

Downloads

Files

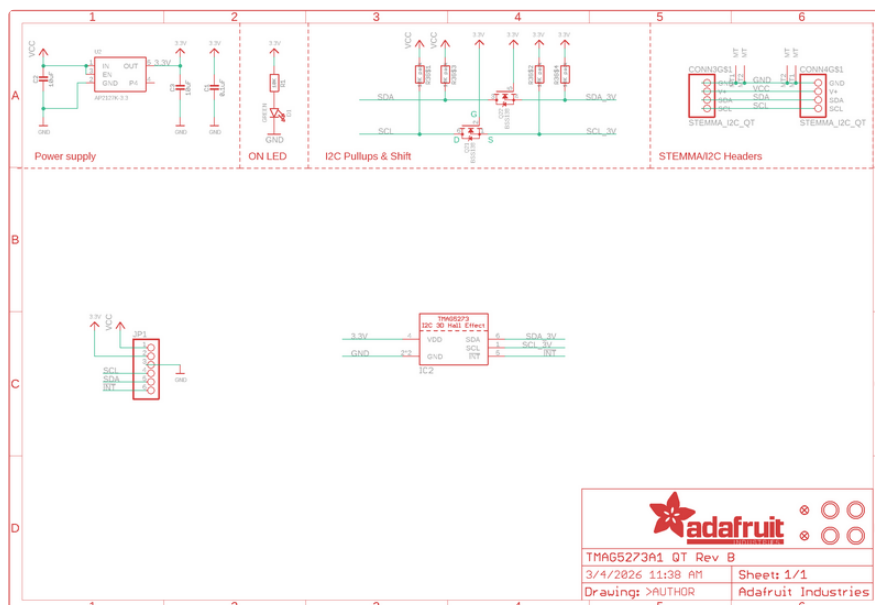
- [TMAG5273 Datasheet \(https://adafru.it/1aC1\)](https://adafru.it/1aC1)
- [EagleCAD PCB files on GitHub \(https://adafru.it/1aC2\)](https://adafru.it/1aC2)
- [Fritzing object in the Adafruit Fritzing Library \(https://adafru.it/1aC3\)](https://adafru.it/1aC3)

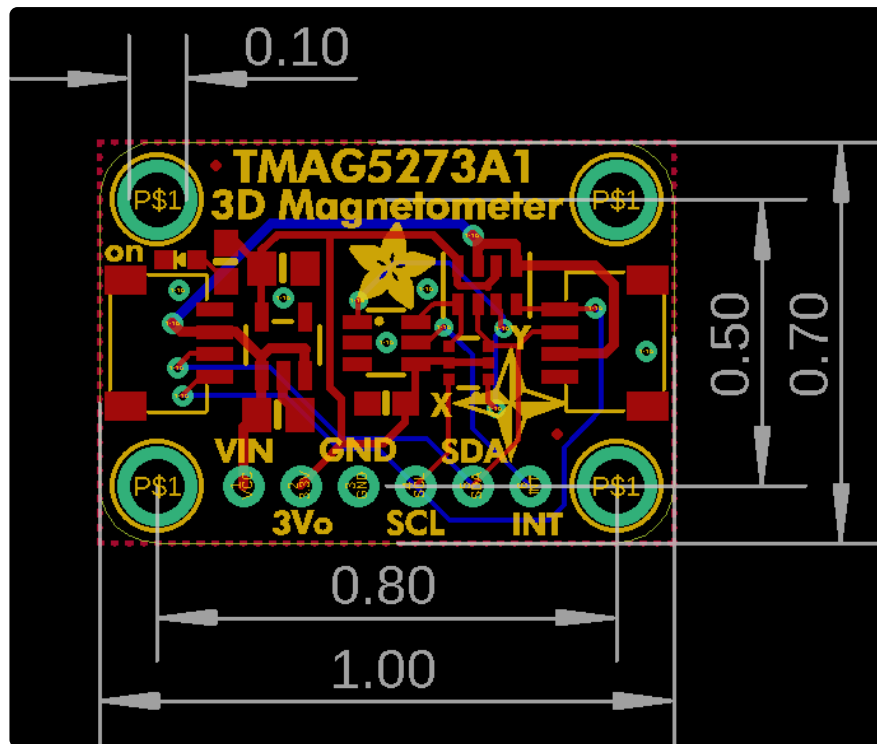
Schematic and Fab Print



A1 and A2 variants have identical schematics and board layouts. The only difference is the silkscreen.

A1





A2

